

- **Thinger.io:** This is an open source platform for IoT and provides scalable cloud infrastructure for connecting devices. It supports all types of boards like Arduino, ESP8266, Raspberry Pi and Intel Edison.
- **Open source IoT hardware - development boards:** Various hardware development boards like Arduino, Freeduino, Raspberry Pi, BBC MicroBit, Orange Pi, Pine A64 as well as operating systems like Raspbian, Kodi, etc, are based on open source.

5. Open source and Big Data analytics

IDC says that worldwide revenues for Big Data and business analytics will be up from US\$ 130 billion in 2016 to more than US\$ 203 billion in 2020 with an annual growth rate of 11.7 per cent. Nowadays, most organisations understand the value of capturing all the data streaming inside the business and hence employ open source Big Data analytics to gain crucial advantage from it.

Open source software and Big Data go hand in hand these days since today's applications can handle diverse data in an effective manner, as it grows exponentially in variety, volume, velocity and veracity.

The popular Big Data analytics tools and platforms are:

- **Apache Hadoop:** Low cost distributed computing for Big Data
- **Grid Gain:** Faster analysis of real-time data
- **Cassandra:** Manages huge databases
- **Terrastore:** Popular for scalability and elasticity
- **KNIME:** Best tool for performance management and data integration
- **Rapidminer:** For faster processing of data and to simplify predictive analysis
- **Solr:** Scalable and reliable tool for Big Data file transfer and aggregation
- **Terracotta:** Enables enterprise applications to store and manage Big Data in server memory
- **AVRO:** Data serialisation system based on JSON-defined schemas
- **Oozie:** Coordinates scheduling of Hadoop jobs
- **Zookeeper:** Centralised service for maintaining configuration information, naming, distributed synchronisation and group services

6. Progressive Web Apps (PWAs)

Progressive Web Apps (PWAs) bring a mobile-app like experience to end users without any app installation requirements. Designed by Google, these apps were promoted in the Google I/O 2017 conference.

PWAs take advantage of the much larger Web ecosystem, plugins, community and the relative ease of deploying and maintaining a website when compared to a native application in the respective app stores. A PWA takes advantage of a mobile app's characteristics, resulting in improved user retention and performance, without the

complications involved in maintaining a mobile application.

Progressive websites are rapidly growing in popularity — as a way to build apps with JavaScript, CSS and HTML, and they have a level of performance and usability that's nearly identical to native apps.

PWAs are able to work with most browsers and devices, fit in all screens with responsive designs, enable offline connectivity, and offer an app-like experience with features like push notifications and Web app manifest.

Tools for building Progressive Web Apps

- **React:** Managed and supported by Facebook and is the foundation for React Native. It can easily port apps built with React to native apps.
- **Polymer Template:** Supported by Google to use the PRPL pattern to optimise delivery of the app to the device.
- **Webpack:** Essential for complex and front-end driven progressive websites and comprises innumerable JavaScript applications.
- **Lighthouse:** Powerful Google PWA performance monitoring tool to test load times and performance in page loads, as well as security in network connections, design and the user interface.

7. Open source and cyber security

When creating a security policy for any organisation, or when building a security operations or research centre, the prime requirement is to have the right people, processes and effective tools.

The open source market is filled with lots of security tools and even Linux distributions like Kali Linux, the Parrot Security Tool Set, Network Security Tool Kit, Cyborg Hawk and many more. All these are used by security and penetration testing professionals for real-time security operations and to find vulnerabilities and backdoors in existing networks.

Open source intelligence (OSINT) is a methodology for using open source tools to collect information from publicly available sources, carry out analysis on the data and take the right course of action. To prevent security attacks on a network, it is of utmost importance to understand the information being collected by the organisation and the software being used to collect it.

As per the latest open source security analysis reports (Blackduck Software-www.blackducksoftware.com), 67 per cent of the applications reviewed contain OS security vulnerabilities, with 40 per cent of open source vulnerabilities in each application being rated as 'severe'.

In the US, it is expected that the 2018 budget for the Pentagon and the US' Department of Defense (DoD) will include the launch of a new pilot programme, for which 20 per cent of the custom code developed will use open source software because as per Linus's Law: "Open source software is more secure as compared to anything else." GitHub hosts more than 70 million open source projects, of which 600,000 components are downloaded over 14 billion times.